Syntax-prosody alignment produced by people and Alexa in conversational speech

Traditionally, it has been assumed that syntactic boundaries are consistently marked by prosodic boundaries in spoken language [1]. However, recent findings have suggested that the syntax-prosody alignment in human speech is not as strong as previously thought [2-3]. However, these findings are limited to non-interactive speech (i.e., with no addressee). Greater alignment between syntactic and prosodic boundaries may be a facet of clear speech and thus may be constrained to more specific audiences, like child-directed speech or when talking to an AI device, in an effort to increase intelligibility. At the same time, the extent to which text-to-speech (TTS) voices produce syntax-prosody alignment is understudied, but one possibility is that it may be stronger due to punctuation in text (e.g., more pauses after a syntactic phrase in [4]). To our knowledge, no prior work has investigated syntax-prosody alignment at the word-level by both people and TTS voices during Human-computer interaction.

To investigate this, we look at unscripted conversations (mean 11.3 min) between adult participants (M=5, F=8 mean age=20) and the Alexa socialbot (engaged by saying “Let’s chat”). Adopting methodology from [2], transcriptions were hand-corrected and then passed through the Montreal Forced Aligner to get the text-to-speech alignment for both the participant and Alexa. We then hand-corrected the resulting TextGrids at the word-level. Next, we used the pretrained pipeline spaCy with the Berkeley Neural Parser to extract syntactic dependencies and identify syntactic boundaries. Simultaneously, to identify prosodic boundaries, we used the Wavelet Prosody Toolkit [5] to extract prosodic data (pitch, duration, intensity) from the speech signal and align it with each word. Upon getting the syntactic and prosodic information, we analyzed the data by looking at the word-level prosodic and syntactic information. For the syntactic information, we looked at whether each word was associated with any closing node (syntactic boundary) or more than two closing nodes (strong syntactic boundary), hypothesizing that strong syntactic boundaries would lead to better syntax-prosody alignment.

Our results show a similar pattern in both human and Alexa productions. At any syntactic boundaries, there is a high level of misalignment for both, with a majority of syntactic boundaries not being marked by a prosodic boundary (Figures 1 & 3). However, increasing the syntactic boundary strength improves alignment, with strong syntactic boundaries marked more frequently by prosodic boundaries in both human and Alexa speech (Figures 2 & 4). We used one-tailed binomial test to determine whether the likelihood that a syntactic boundary also contained a prosodic boundary was greater than chance; in human speech, the success probability was 0.70 ($p < 0.001$), 95% CI [0.66,1]), while in Alexa speech, it was was 0.77 ($p < 0.001$), 95% CI [0.75,1]). Both of these effects were relatively large (Cohen’s h=0.41 and 0.56, resp., for human and Alexa speech).

Our results suggest that human and Alexa speech exhibit similar patterns of syntax-prosody (mis)alignment in a conversational context. Specifically, they produce prosodic adjustments on the final word of a strong syntactic phrase. However, in weaker syntactic phrases, no alignment is observed, consistent with work showing that syntactic-prosodic alignment is not always robust [2-3]. Here, one possibility is that the Alexa TTS voice generated stronger syntactic-prosodic alignment, both based on commas in the transcribed text and the lack of a need for breathing, but that producing no prosodic boundaries might also disrupt the natural rhythm of speech [6]. With *human* speech patterning similarly, this could be a characteristic of clear speech and/or mirroring of the Alexa pattern. Likewise, the higher alignment in Alexa speech could contribute to Alexa’s unnaturalness.

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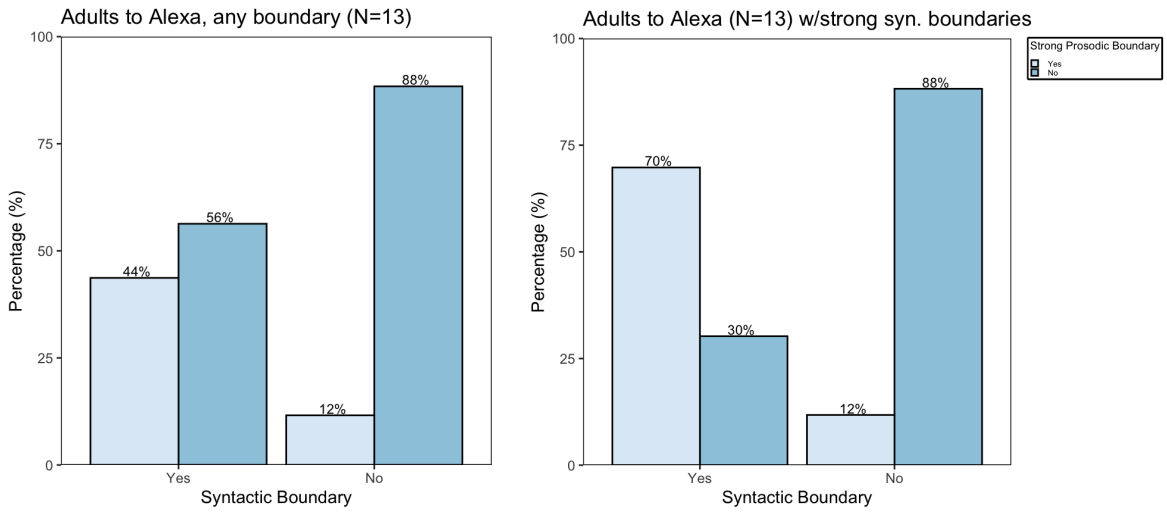
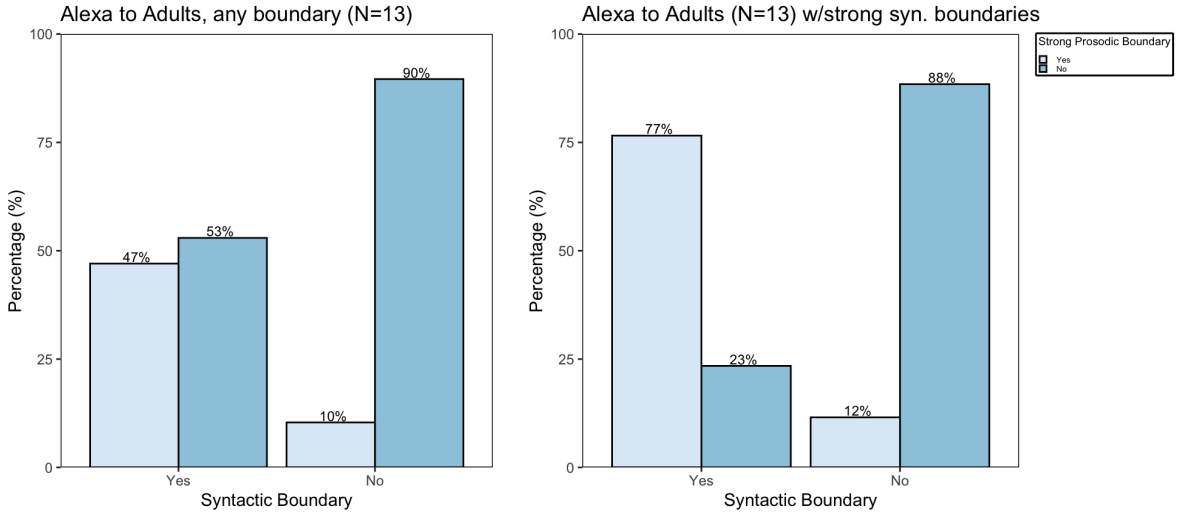


Figures 1 (left) and 2 (right) : Syntax-prosody alignment in human speech, any syntactic boundaries (left) and strong syntactic boundaries (right)



Figures 3 (left) and 4 (right) : Syntax-prosody alignment in Alexa socialbot speech, any syntactic boundaries (left) and strong syntactic boundaries (right)

References :

[1] Bennett, R., & Elfner, E. (2019). The syntax–prosody interface. Annual Review of Linguistics, 5(1), 151-171.
[2] Degano, G., Donhauser, P. W., Gwilliams, L., Merlo, P., & Golestani, N. (2024). Speech prosody enhances the neural processing of syntax. Communications Biology, 7(1), 748.
[3] ANONYMIZED. (submitted). Syntactic and prosodic phrasal alignment in naturalistic language.
[4] Chan, C., & Kuang, J. (2024). Exploring the accuracy of prosodic encodings in state-of-the-art text-to-speech models. In Proceedings Speech Prosody (Vol. 2024, pp. 27-31).
[5] Suni, A., Šimko, J., Aalto, D., & Vainio, M. (2017). Hierarchical representation and estimation of prosody using continuous wavelet transform. Computer Speech & Language, 45, 123-136.
[6] Inbar, M., Grossman, E., Landau, A.N.. (2025). A universal of speech timing: Intonation units form low-frequency rhythms, Proc. Natl. Acad. Sci. U.S.A. 122 (34) e2425166122.